

Technical Note

1. Introduction

This project presents a lightweight decision-support framework for unmanned aerial vehicle (UAV) cargo operations in a coastal urban environment. The objective was not only to generate synthetic operational data but also to transform those data into meaningful performance indicators that can support mission planning and post-flight evaluation. The solution was developed in three stages: synthetic dataset generation, rule-based safety classification and mission performance scoring. The final outcome is a reproducible workflow capable of comparing multiple missions under different operational priorities.

2. Dataset Generation

Five representative missions were generated, producing more than 200 flight records sampled every 30 seconds. Each mission follows a realistic flight profile consisting of climb, cruise and descent phases. Environmental variables such as wind speed, wind gusts, visibility, humidity and temperature evolve coherently instead of changing randomly. Battery consumption, speed and acoustic emissions are correlated with payload and weather conditions to maintain physically plausible behaviour. Two special scenarios were deliberately included: an emerging storm mission and a battery-intensive mission caused by heavy payload and strong wind.

2.1 Mission Profiles

Mission	Scenario	Purpose	Main Challenge
Mission 1	Ideal Conditions	Baseline reference	None
Mission 2	Good Conditions	Payload evaluation	Higher battery consumption
Mission 3	Emerging Storm	Weather robustness	Increasing wind and decreasing visibility
Mission 4	High Battery Drain	Operational stress test	Heavy payload and strong wind
Mission 5	Emergency Delivery	Rapid response	Short mission optimisation

Mission 1 represents the baseline flight under ideal weather. Mission 2 investigates the influence of heavier payload while maintaining favourable environmental conditions. Mission 3 introduces progressively worsening weather, allowing the safety classifier to respond naturally to increasing operational risk. Mission 4 isolates battery degradation caused primarily by heavy payload and strong sustained wind rather than poor visibility. Mission 5 represents a short emergency delivery mission focused on rapid deployment.

3. Safety Classification

Each flight segment is evaluated using configurable thresholds stored externally in a JSON configuration file. The classifier assigns every segment to one of three operational states: safe, degraded or unsafe. Unsafe conditions always have priority, while multiple degraded conditions may escalate into an unsafe classification. Separating thresholds from the implementation makes the framework easily configurable without changing the source code.

4. Mission Performance Score

Mission performance is represented through three normalized indices: the Environmental Efficiency Index (EEI), Operational Efficiency Index (OEI) and Noise Impact Index (NII). Each index is normalized between 0 and 1 before being combined into the Mission Performance Score (MPS). Three weighting strategies were evaluated: Balanced, Environment Priority and Noise Priority. Comparing rankings obtained from these configurations highlights how operational priorities influence mission assessment.

5. Results and Discussion

Mission 1 consistently achieved the highest performance regardless of weighting strategy, indicating balanced environmental conditions and efficient resource utilisation. Mission 4 consistently ranked last because its operational penalties outweighed its remaining strengths. Missions 2 and 5 exchanged positions depending on the selected weighting strategy, making them the most sensitive missions within the dataset. Two complementary figures were generated: one comparing mission rankings under different weighting configurations and another illustrating the weighted contribution of EEI, OEI and NII to the final Mission Performance Score.

5.1 Generated Outputs

The software automatically produces five deliverables: `flight_log.csv`, `mission_summary.csv`, `missions_score.csv`, `mission_performance_comparison.png` and `mission_scores.png`. These outputs provide both numerical analysis and graphical interpretation of mission behaviour, making the framework suitable for post-flight evaluation.

6. Limitations

The proposed framework intentionally simplifies several real-world phenomena. Weather conditions are synthetic, battery ageing is not modelled explicitly and aircraft dynamics are represented through simplified empirical relationships rather than detailed aerodynamic simulations. Nevertheless, these simplifications were considered appropriate because the objective of the task was to demonstrate software architecture, data processing and decision-support concepts rather than develop a certified flight model.

7. Future Development

The current implementation provides a solid foundation for future work. A natural extension would be replacing synthetic weather generation with historical or forecast data retrieved automatically from a public weather API, allowing missions to be simulated under realistic conditions for any location and season. Additional mission categories could also be introduced, including light cargo, medium cargo, heavy cargo, emergency delivery, reconnaissance, humanitarian assistance, infrastructure inspection and maritime intelligence missions. A particularly interesting application would involve coastal surveillance drones equipped with thermal cameras capable of detecting approaching vessels and supporting visual confirmation of vessel identity. The scoring framework could further evolve by introducing adaptive weighting based on mission objectives, additional environmental variables such as precipitation or turbulence, battery ageing models and machine-learning techniques for predictive mission assessment.

8. Conclusion

The developed solution successfully satisfies the technical objectives by integrating synthetic data generation, configurable safety assessment, mission performance scoring and visual analytics into a single workflow. The modular implementation allows future extensions with minimal changes, making the framework suitable not only as a technical exercise but also as the basis for more advanced UAV mission planning and operational decision-support systems.